Trainings

Select Service Tools

Recommended Cummins® Service Tools

- Exhaust Gas Recirculation (EGR) Cooler Leak Test Kit, Part Number 5573379
- Air Pressure Regulator Kit, Part Number 3164231, or equivalent

Additional Service Items

- Protective caps, or equivalent
- High-temperature anti-seize compound.

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : Brush away all loose dirt from around the area of the air handling connections to reduce the possibility of contamination of the interior of the engine.

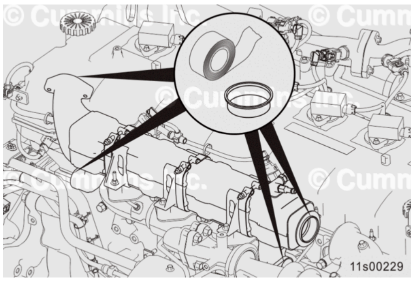
Note : If the engine is equipped with Closed Crankcase Ventilation (CCV) and the EGR cooler is being replaced due to a malfunction that causes an internal coolant leak, the crankcase breather element **must** be changed. See equipment manufacturer service information.

- Drain the engine coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Remove the EGR crossover tube. Refer to Procedure 011-070 in Section 11. (/qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html)

- Remove the air compressor inlet tube. Refer to Procedure 012-109 in Section 12. (/qs3/pubsys2/xml/en/procedures/1016/1016-012-109.html)
- Remove the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/1016/1016-011-031.html)
- Remove the EGR cooler connection. Refer to Procedure 011-024 in Section 11. (/qs3/pubsys2/xml/en/procedures/1016/1016-011-024.html)

Remove

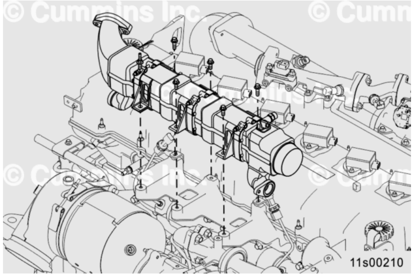
Use protective caps, or equivalent, to cover the open points on both the EGR cooler and the EGR plumbing.



Remove the five capscrews and one stud capscrew that attach the EGR cooler to the EGR cooler mounting bracket.

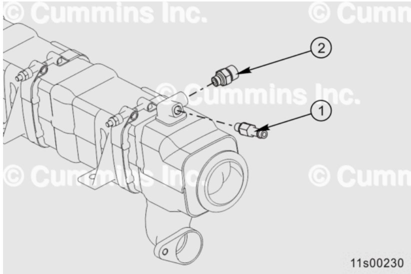
Remove the EGR cooler assembly from the engine.

Note : Carefully remove the EGR cooler and do not damage the two exhaust gas pressure sensor tubes.



Disassemble

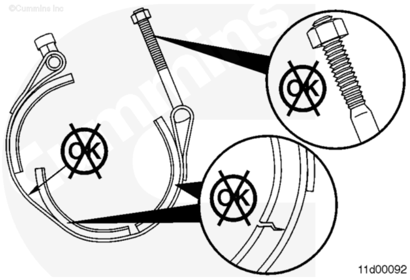
Remove the two connectors (1 and 2) from EGR cooler vent port.



Clean and Inspect for Reuse

Inspect the V-band clamps and EGR bellows for damage. Refer to Procedure 011-024 in Section 11. (/qs3/pubsys2/xml/en/procedures/1016/1016-011-024.html)

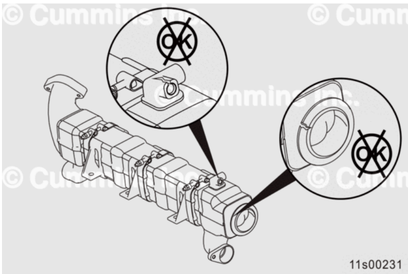
Replace the V-band clamp or EGR bellows if cracks or other damage is found.



Inspect the EGR cooler exhaust gas inlet, outlet, and the mounting surfaces for signs of fretting, cracks, or other damage.

Inspect the EGR cooler or connectors for cracks or other damage.

Replace the EGR cooler or connectors if cracks or other damage is found.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

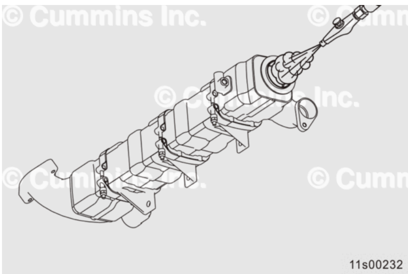
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the EGR cooler exhaust gas inlet, outlet, and the mounting surfaces with safety solvent to remove any coolant deposits or debris.

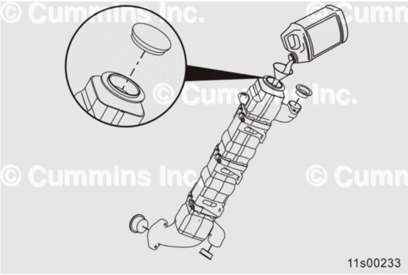
Clean the EGR cooler coolant supply and return tubes to remove any coolant deposits or debris at the o-ring joint.

Dry with compressed air.



⚠ CAUTION ⚠

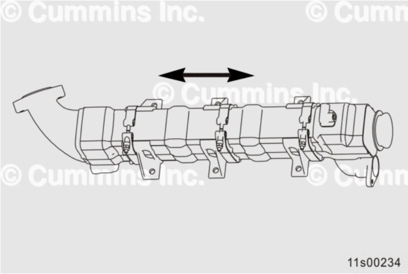
Do not use chlorinated water to clean or rinse the EGR cooler. Chlorides are very corrosive to the stainless alloys internal and external to the EGR cooler.



Cap the exhaust gas outlet of the EGR cooler with the protective cover, or equivalent.

Completely fill the EGR cooler with safety solvent.

Cap the exhaust gas inlet of the EGR cooler with the protective cover, or equivalent.



Lay the EGR cooler on its side.

Allow the EGR cooler to soak for approximately 20 minutes.

Drain approximately 25 percent of the liquid from the cooler and recap. Dispose of the liquid properly.

Shake the cooler by hand, from end to end, for approximately 30 seconds.

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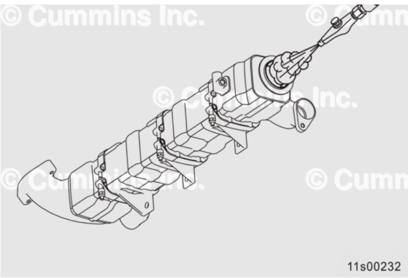


Drain the remaining safety solvent from the EGR cooler and properly dispose of it.

Use compressed air to dry the inside of the EGR cooler and remove any loose debris or particles.

Cap the exhaust gas outlet of the EGR cooler with the protective cover, or equivalent.

Completely fill the EGR cooler with water.

Cap the exhaust gas inlet of the EGR cooler with the protective cover, or equivalent.

Shake the cooler by hand, from end to end, for approximately 30 seconds.

Drain the water from the EGR cooler and properly dispose of it.

Use compressed air to dry the inside of the EGR cooler.

Pressure Test

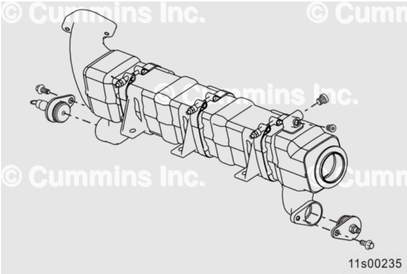
Note : This test is performed with the EGR cooler removed from the engine.

It is necessary to submerge the EGR cooler in warm water to pressure test and check for leaks.

Install the plate with the o-ring supplied in the EGR Cooler Leak Test Kit, Part Number 5573379, into the coolant inlet side of the EGR cooler. Use M8 capscrew.

Install two straight thread plugs from the EGR Cooler Leak Test Kit, Part Number 5573379, into the vent port.

Install the plate onto the coolant outlet provided in the EGR Cooler Leak Test Kit, Part Number 5573379. Use M8 capscrew.



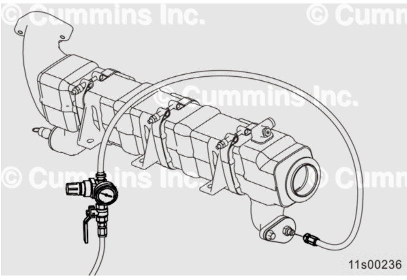
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : Air Pressure Regulator Kit, Part Number 3164231, or equivalent, is suitable for use during this step.

Install the air pressure regulator and hose assembly onto the test fitting on the coolant inlet of the EGR cooler.

Install a ball-type shutoff valve onto the inlet side of the air pressure regulator, to allow the air supply to be shut off during testing.

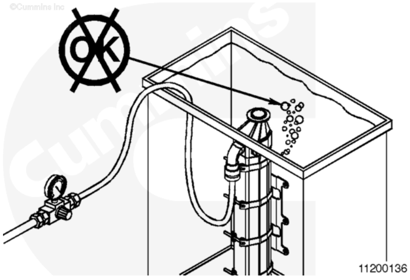


Apply 500 kPa [73 psi] of air pressure to the EGR cooler.

Submerge the EGR cooler vertically into a tank of water. Make sure the cooler is completely submerged.

Note : If vertical orientation is not possible, horizontal orientation is acceptable; however, trapped air may be more difficult to release.

Rotate and shake the cooler, from end to end, to allow any trapped air to escape. Observe the cooler over a 30 second period of time.



Large, slowly forming bubbles do **not** signify a leaking EGR cooler.

Verify that bubbles are **not** a result of loose fittings or trapped air.

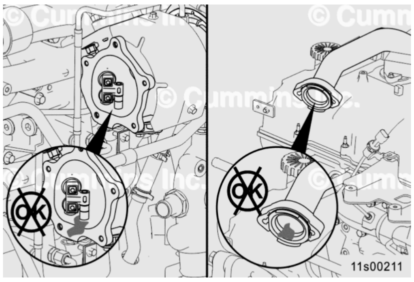
Replace the EGR cooler if a steady stream of bubbles is observed coming from either end of the cooler. If bubbles are **not** observed, the EGR cooler has passed the submerge test.

If the EGR cooler passes the submerge test, the EGR cooler is reusable. Remove the leak test equipment and use compressed air to dry the inside of the EGR cooler.

Note : If a steady stream of bubbles is seen, carefully inspect the air pressure regulator, air lines, ball valves, o-rings, and other components for leaks.

Open the ball valve to release any pressure in the EGR cooler before removing the test fittings.

If the EGR cooler is **not** reusable, inspect the turbocharger turbine housing and rear EGR connection tube for any signs of coolant deposits.



Coolant deposits can be identified as dried white deposits coating the inside of the turbine housing.

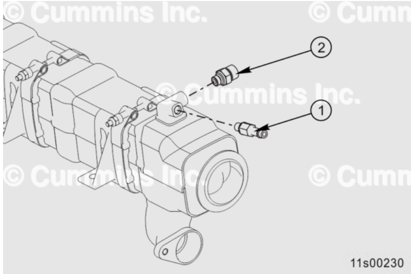
If coolant deposits are found in the turbocharger turbine housing, clean the turbine housing and inspect the wastegate. Also, clean and inspect the EGR differential pressure sensor and the EGR crossover tube.

See the following procedures:

- Turbocharger: Refer to Procedure 010-033 in Section 10 (/qs3/pubsys2/xml/en/procedures/1016/1016-010-033.html).
- EGR differential pressure sensor: Refer to Procedure 019-370 in Section 19. (/qs3/pubsys2/xml/en/procedures/1016/1016-019-370.html)
- EGR crossover tube: Refer to Procedure 011-070 in Section 11. (/qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html)

Assemble

Install the two connectors (1 and 2) to EGR cooler vent port.

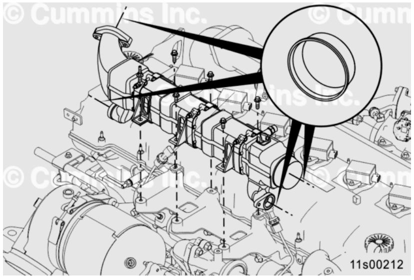


Torque Value:
Connector (1) 24 n•m   [212 in-lb]

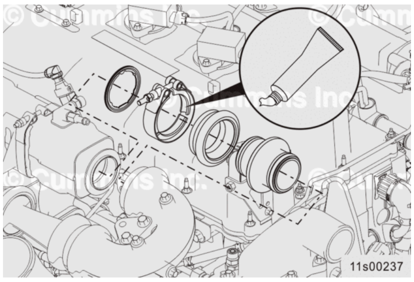
Torque Value:
Connector (2) 34 n•m   [25 ft-lb]

Install

Remove the protective caps, or equivalent, used to cover the open ports on the EGR cooler.



Apply a film of high-temperature anti-seize compound to the inside of the V-band clamp to make sure of proper loading on the clamp.

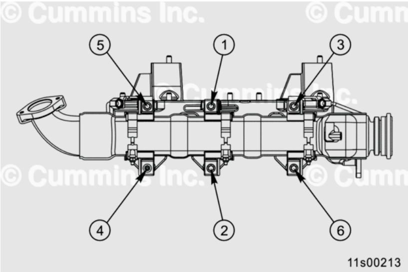


Install a V-band clamp onto the EGR cooler bellows.

Install a new gasket on the pilot end of the EGR cooler, where it connects to the EGR cooler bellows.

Install the EGR cooler on the studs in the lubricating oil cooler assembly.

Install the five EGR cooler mounting capscrews and one stud capscrew.



Tighten the capscrews in a star pattern.

Torque Value: 24 n•m   [212 in-lb]

Finishing Steps

- Install the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/1016/1016-011-031.html)
- Install the EGR cooler connection. Refer to Procedure 011-024 in Section 11. (/qs3/pubsys2/xml/en/procedures/1016/1016-011-024.html)
- Install the air compressor inlet tube. Refer to Procedure 012-109 in Section 12. (/qs3/pubsys2/xml/en/procedures/1016/1016-012-109.html)
- Install the EGR crossover tube. Refer to Procedure 011-070 in Section 11. (/qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html)
- Fill the engine cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Operate the engine to check for leaks. Verify that all fault codes are inactive.

Last Modified: 14-Dec-2022